Bryant Harr

Note: $n_i = 10^{10}$ is used instead of $1.45 \cdot 10^{10}$ per footnote 53

Q1 (1.1)

a) By table 1.1, $\mu_n = 1194 \frac{cm^3}{V \cdot s}$

$$J_n = n q \mu_n E \Rightarrow \sigma = n q \mu_n \Rightarrow \rho = \frac{1}{1194 \cdot 10^{16} \cdot 1.6 \cdot 10^{-19}} = \boxed{\rho = 0.523\ \Omega \cdot cm}$$

$\frac{1}{1.9104}$

b) By table 1.3, Si: $5 \cdot 10^{22} / cm^3$ @ 28 weight. P: $10^{16} / cm^3$ @ 31 weight

$$\frac{P}{Si} = \frac{31 \cdot 10^{16}}{28 \cdot 5 \cdot 10^{22}} = \boxed{2.214 \cdot 10^{-7} = \text{P weight proportion}}$$

(technically should be $\frac{P}{P+Si}$, but P is so small, is basically same as $\frac{P}{Si}$)

c) $\underbrace{10^{17}}_{B} - \underbrace{10^{16}}_{P} = p = 9 \cdot 10^{16}\ 1/cm^3 > 0 \Rightarrow$ p-type material (More Boron)

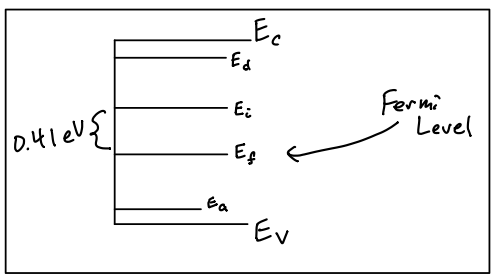
By fig 1.16 at Boron curve, $\mu_p = 300 - 320 \approx 310 \frac{cm^3}{V \cdot s}$

As before, $J_p = p q \mu_p E \Rightarrow \sigma = p q \mu_p = 4.464 \quad \rho = \frac{1}{\sigma} = 0.224$

$\boxed{\text{p-type material}}$ $\boxed{\rho = 0.224\ \Omega \cdot cm}$ (± 0.01) 300-320

d) From 1.1.27, $p = n_i \exp\left(\frac{E_i - E_f}{kT}\right) \Rightarrow E_i - E_f = kT \ln\left(\frac{p}{n_i}\right) = 0.026 \ln\left(\frac{9 \cdot 10^{16}}{1.45 \cdot 10^{10}}\right) = 0.41\ eV = E_i - E_f$

$\hookrightarrow 1.45 \cdot 10^{10}$ by table 1.3



Q2 (1.3)

(As) $n \approx N_d = 10^{16}\ 1/cm^3$ (B) $p \approx N_a = 10^{15}\ 1/cm^3$

From Table 1.3, $N_c = 2.8 \cdot 10^{19}\ \frac{1}{cm^3}$ $N_v = 1.04 \cdot 10^{19}\ \frac{1}{cm^3}$ (@ 300K)

From below Table 1.4, Arsenic: $E_c - E_d \approx 0.049eV$ Boron: $E_a - E_v \approx 0.045eV$

(All constants I invoke come from same table/figure + rounding

a) Arsenic: Eq 1.1.21: $n = N_c \exp(-\frac{E_c - E_f}{kT})$. Note: $N_c = N_{c300}(\frac{T}{300})^{3/2}$ (@ 300K)

Sample

Half ionized $\Rightarrow n = \frac{N_d}{2} \Rightarrow \frac{N_d}{2} = N_c \exp(-\frac{\overset{0.049}{E_c - E_f}}{kT}) \Rightarrow \ln(\frac{N_d}{2N_{c300} \cdot (T/300)^{3/2}}) = -\frac{0.049}{kT}$

$\rightarrow \ln(\frac{N_d}{2N_{c300} \cdot (T/300)^{3/2}}) = -\frac{0.049}{kT} \Rightarrow \ln(5600(T/300)^{3/2}) = 568.62/T$.

$\ln(5600(T/300)^{3/2}) = 568.62/T \Rightarrow$ Solving by graphing $\Rightarrow$ $\boxed{T \approx 84.4°K}$ for Arsenic sample

84-85°

Boron:

As before, same process. $p = N \exp(-\frac{E_a - E_v}{kT})$ Note: $N_v = N_{v300}(\frac{T}{300})^{3/2}$ (@ 300K)

$p = \frac{N_a}{2} \Rightarrow \frac{N_a}{2} = N_v \exp(-\frac{E_a - E_v}{kT}) \Rightarrow \ln(\frac{N_a}{2N_{v300} \cdot (T/300)^{3/2}}) = -\frac{0.045}{kT} \Rightarrow \ln(20800(\frac{T}{300})^{3/2}) = 522.2/T$

$\ln(20800(\frac{T}{300})^{3/2}) = 522.2/T \Rightarrow$ Solving by graphing $\Rightarrow$ $\boxed{T \approx 67.7°K}$ for Boron sample

b) $n_i = 3.87 \cdot 10^{16} T^{\frac{3}{2}} \exp(-\frac{7014}{T})$

As: $n_i = 10 \cdot \underbrace{10^{16}}_{N_d} = 10^{17} \Rightarrow 10^{17} = 3.87 \cdot 10^{16} T^{\frac{3}{2}} e^{-\frac{7014}{T}} \Rightarrow$ Solve for T by graphing $\Rightarrow$ $\boxed{T \approx 777°K}$ for Arsenic sample

B: $n_i = 10 \cdot \underbrace{10^{15}}_{N_d} = 10^{16} \Rightarrow 10^{16} = 3.87 \cdot 10^{16} T^{\frac{3}{2}} e^{-\frac{7014}{T}} \Rightarrow$ Solve for T by graphing $\Rightarrow$ $\boxed{T \approx 635°K}$ for Boron sample

c) As: $np = n_i^2 = 10^{16} p = (2.1 \cdot 10^{10})^2 \Rightarrow$ $\boxed{p = 2.1 \cdot 10^4\ cm^{-3}}$ for Arsenic sample

B: $np = n_i^2 = 10^{15} n = (2.1 \cdot 10^{10})^2 \Rightarrow$ $\boxed{n = 2.1 \cdot 10^5\ cm^{-3}}$ for Boron sample

d) $E_i - E_v = 0.56eV$

As: $E_f - E_i = -kT \ln(\frac{n}{n_i}) = 0.026 \ln(\frac{10^{16}}{1.45 \cdot 10^{10}}) = 0.35eV$

$\Rightarrow 0.56 + 0.35 = \boxed{E_f - E_v = 0.91\ eV}$ for Arsenic

B: $E_f - E_i = -kT \ln(\frac{p}{n_i}) = -0.026 \ln(\frac{10^{15}}{1.45 \cdot 10^{10}}) = -0.29eV$

$\Rightarrow E_f - E_v = -0.29 + 0.56 = \boxed{E_f - E_v = 0.27\ eV}$ for Boron

As+B: $E_f - E_i = kT \ln(\frac{\Delta n, p}{n_i}) = 0.026 \ln(\frac{\overset{\rightarrow N_d - N_a}{10^{16} - 10^{15}}}{1.45 \cdot 10^{10}}) = 0.3468eV$

$E_f - E_v = 0.3468 + 0.56 \approx \boxed{0.907\ eV}$ for both As + B

Q3 (1.11)

a) $n_d = 0.05 \cdot 10^{10} = 5 \cdot 10^8$ $\boxed{\text{Donor: } 5 \cdot 10^8 / cm^3}$ (Sub Gallium) - donor: Si has 1 more electron

$n_a = 0.95 \cdot 10^{10} = 9.5 \cdot 10^9$ $\boxed{\text{Acceptor: } 9.5 \cdot 10^9 / cm^3}$ (Sub Arsenic) - acceptor:

b) $p = n_a - n_d = 9 \cdot 10^9 / cm^3$ By table 1.3 GaAs @ 300K: $n_i = 9 \cdot 10^6$, $N_v = 7 \cdot 10^{18}$

$np = n_i^2 \Rightarrow n = \frac{n_i^2}{p} = \frac{81 \cdot 10^{12}}{9 \cdot 10^9} = 9 \cdot 10^3 / cm$ $E_f - E_v = kT \ln\left(\frac{N_v}{p}\right) = 0.026 \cdot \ln\left(\frac{7 \cdot 10^{18}}{9 \cdot 10^9}\right) = 0.5322$

$\Rightarrow \boxed{p = 9 \cdot 10^9 \frac{1}{cm^3}}$ $\boxed{n = 9 \cdot 10^3 \frac{1}{cm^3}}$ $\boxed{E_f = 0.53 eV \text{ over } E_v}$

c) As before for only p, $J_p = p q_h \mu_p E \Rightarrow \sigma = p q_h \mu_p$. Analogously for both, $\sigma = n q_h \mu_n + p q_h \mu_p$

By Table 1.3, $\mu_n = 8800$, $\mu_p = 400$ $\Rightarrow \sigma = n q_h \mu_n + p q_h \mu_p = 8800 \cdot 1.6 \cdot 10^{-19} \cdot 9 \cdot 10^3 + 400 \cdot 1.6 \cdot 10^{-19} \cdot 9 \cdot 10^9 = 5.76 \cdot 10^{-7}$

$\boxed{\sigma = 5.76 \cdot 10^{-7} \frac{1}{\Omega \cdot cm}}$ $\left(\frac{S}{cm}\right)$

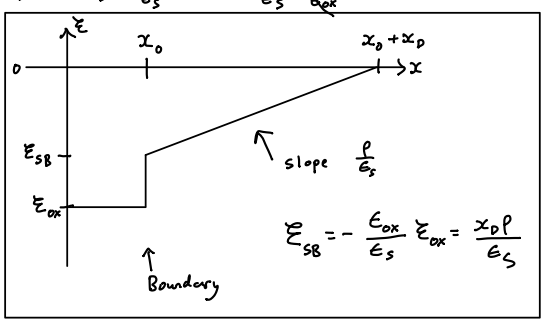
Q4 (4.1.4)

a) @ $x=0$, balancing charge $= -x_D \rho$. At $x<0$, metal $\Rightarrow \mathcal{E} = 0$. (0 field)

In oxide, $\mathcal{E}_{ox} = \frac{\text{charge}}{\epsilon} = -\frac{x_D \rho}{\epsilon_{ox}}$. Call oxide $\mathcal{E}_{ox}$, semi field $\mathcal{E}_s$, fields @ boundary $\mathcal{E}_{oxB}$, $\mathcal{E}_{sB}$.

Note $\mathcal{E}_{ox} = \mathcal{E}_{oxB}$, so we use $\mathcal{E}_{ox} \Rightarrow$ At boundary, $\mathcal{E}_{ox} \epsilon_{ox} = \mathcal{E}_s \epsilon_{sB} \Rightarrow \mathcal{E}_{sB} = \frac{\epsilon_{ox}}{\epsilon_s} \mathcal{E}_{ox} = -\frac{\cancel{\epsilon_{ox}}}{\epsilon_s} \frac{x_D \rho}{\cancel{\epsilon_{ox}}} \Rightarrow \mathcal{E}_s = \mathcal{E}_{sB} + \frac{\rho(x - x_0)}{\epsilon_s}$

$$\Rightarrow \boxed{\mathcal{E} = \begin{cases} -\frac{x_D \rho}{\epsilon_{ox}} & 0 \le x < x_0 \\ -\frac{x_D \rho}{\epsilon_s} + \frac{\rho(x - x_0)}{\epsilon_s} & x_0 < x < x_0 + x_D \\ 0 & \text{otherwise (outside)} \end{cases}}$$



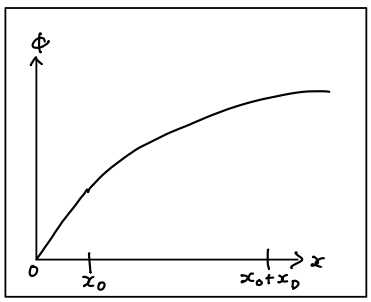
b) Let $\boxed{\mathcal{E}_{ox} = -\frac{\rho x_D}{\epsilon_{ox}}}$ as before

For $x < 0$, $\phi = 0$ (no field) For $0 \le x < x_0$, $\phi_{ox} = \int_0^x \mathcal{E}_{ox} \, dx = \mathcal{E}_{ox} x$

For $x_0 < x < x_0 + x_D$, $\phi_s = \phi_{ox} + \int_{x_0}^x -\frac{x_D \rho}{\epsilon_s} + \frac{\rho(x - x_0)}{\epsilon_s} \, dx = \phi_{ox} + \frac{\rho(x - x_0)^2}{2\epsilon_s} + \left(\cancel{-\frac{x_D \rho}{\epsilon_s}}\right) - \left(\cancel{-\frac{x_D \rho}{\epsilon_s}}\right) = \phi_{ox} + \frac{\rho(x - x_0)^2}{2\epsilon_s}$

For $x > x_0 + x_D$, flattens.

$$\Rightarrow \boxed{\phi = \begin{cases} 0 & x < 0 \\ \mathcal{E}_{ox} x & 0 \le x < x_0 \\ \mathcal{E}_{ox} x_0 + \frac{\rho(x - x_0)^2}{2\epsilon_s} & x_0 \le x < x_0 + x_D \end{cases}}$$



Basically the integral of the above graph

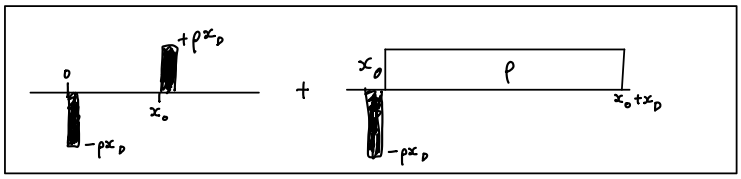
c) By above expression, $\phi(0)=0$, $\phi(x_0) = \mathcal{E}_{ox}x_0 \Rightarrow \Delta\phi_0 = \mathcal{E}_{ox}x_0$

Recall $\mathcal{E}_{ox} = -\frac{\rho x_D}{\epsilon_{ox}} \Rightarrow \boxed{\Delta\phi_0 = -\frac{\rho x_D}{\epsilon_{ox}} x_0}$

d) $\Delta\phi_D = \cancel{\mathcal{E}_{ox}x_0} - \cancel{\mathcal{E}_{ox}x_0} - \frac{\rho(x-x_0)^2}{2\epsilon_s} = \boxed{\Delta\phi_D = \frac{\rho(x-x_0)^2}{2\epsilon_s}}$

e)

Use this decomposition →



$\mathcal{E}_{ox}$ is clearly $-\frac{\rho x_D}{\epsilon_{ox}}$ by first one.

Semi section is just a block with density ρ

$\Rightarrow \frac{\rho(x-x_0)}{\epsilon_s} \Rightarrow$ By same logic as before,

Integrate to get ϕ

$$\boxed{\mathcal{E} = \begin{cases} -\frac{x_D\rho}{\epsilon_{ox}} & 0 \le x < x_0 \\ -\frac{x_D\rho}{\epsilon_s} + \frac{\rho(x-x_0)}{\epsilon_s} & x_0 < x < x_0 + x_D \\ 0 & \text{otherwise (outside)} \end{cases}}$$

$$\boxed{\phi = \begin{cases} 0 & x < 0 \\ \mathcal{E}_{ox}x & 0 \le x < x_0 \\ \mathcal{E}_{ox}x_0 + \frac{\rho(x-x_0)^2}{2\epsilon_s} & x_0 \le x < x_0 + x_D \\ \mathcal{E}_{ox}x_0 + \frac{\rho x_D^2}{2\epsilon_s} & x_0 + x_D \le x \end{cases}}$$